gateTunnelManagement

Service Description

Service ID: *"gateTunnelManagement"*

Abstract
This document describes the `gateTunnelManagement` service, version 5.2.2, produced by the GateTunnel support system of Arrowhead (documentation edition 5.3.0).

Contents

1 Documentation edition and implementation scope **3**

2 Overview **3**
2.1 Significant Prior Art . . . 3
2.2 How This Service Is Meant to Be Used . . . 3
2.3 Important Delimitations and Dependencies . . . 3

3 Service Interface **4**
3.1 operation sessions . . . 4
3.2 operation close . . . 4
3.3 operation status . . . 4

4 Information Model **5**
4.1 struct CloseResult . . . 5
4.2 struct SessionCloseRequest . . . 5
4.3 struct SessionList . . . 5
4.4 struct Status . . . 5
4.5 Primitives . . . 5

5 References **7**

6 Revision History **8**
6.1 Amendments . . . 8
6.2 Quality Assurance . . . 8

1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `gateTunnelManagement` service, which provides operator access to GateTunnel sessions and status. It is produced by the GateTunnel system [1] in service version 5.2.2.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

New in Arrowhead 5.

2.2 How This Service Is Meant to Be Used

LocalCloudManager lists sessions, closes them and shows the relay and session status.

2.3 Important Delimitations and Dependencies

Only LocalCloudManager and Sysop.

3 Service Interface

This section describes the interfaces to the `gateTunnelManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

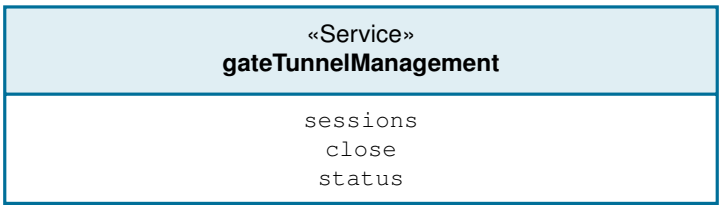


Figure 1: SysML-style block of the `gateTunnelManagement` service and its operations.

The following interface operations are available.

3.1 operation sessions () : SessionList

Returns the control sessions.

3.2 operation close (SessionCloseRequest) : CloseResult

Closes a session by identifier or port.

3.3 operation status () : Status

Returns the GateTunnel status with relay and session state.

4 Information Model

Here, all data objects that can be part of the `gateTunnelManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.5, which are used to represent things like hashes and identifiers.

4.1 struct CloseResult

Field	Type	Description
closed	Boolean	Whether a session was closed; 404 when false.

4.2 struct SessionCloseRequest

Field	Type	Description
sessionId	Identifier	Session to close.
tunnelPort	Number	Or the control port to close.

4.3 struct SessionList

Field	Type	Description
entries	List<Map<String, Object>>	Control sessions: `sessionId`, `queueId`, `peerName`, `serviceDefinition`, `tunnelPort`, `state`, `consumer`, `provider`, `consumerCloud`, `providerCloud`, `relay`, `sessionStartedAt`.
count	Number	Number of entries.

4.4 struct Status

Field	Type	Description
status	String	`ok`.
system	String	`GateTunnel`.
version	Version	`5.2.2`.
implementation	String	`java`.
registration	Map<String, Object>	Registration state.
relay	Map<String, Object>	Relay connection state.
interCloud	Map<String, Object>	`providerSessions`, `consumerSessions` and session summaries.
activeSessions	Number	Control sessions not closed.

4.5 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Number	An integer number.
Object	Any structured value.
String	A chain of characters.
Version	Specifies a version, semantic versioning `major.minor.patch`.

5 References

[1] J. Delsing, “GateTunnel – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-GateTunnel-java-spring/documentation/v5.3.0/GateTunnel-v5.3.0-SysD.pdf.

[2] ——, “gateTunnelManagement HTTP/TLS/JSON – Interface Design Description (GateTunnel),” Eclipse Arrowhead, IDD 5.3.0, 2026, gateTunnel-v5.3.0-gateTunnelManagement-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	